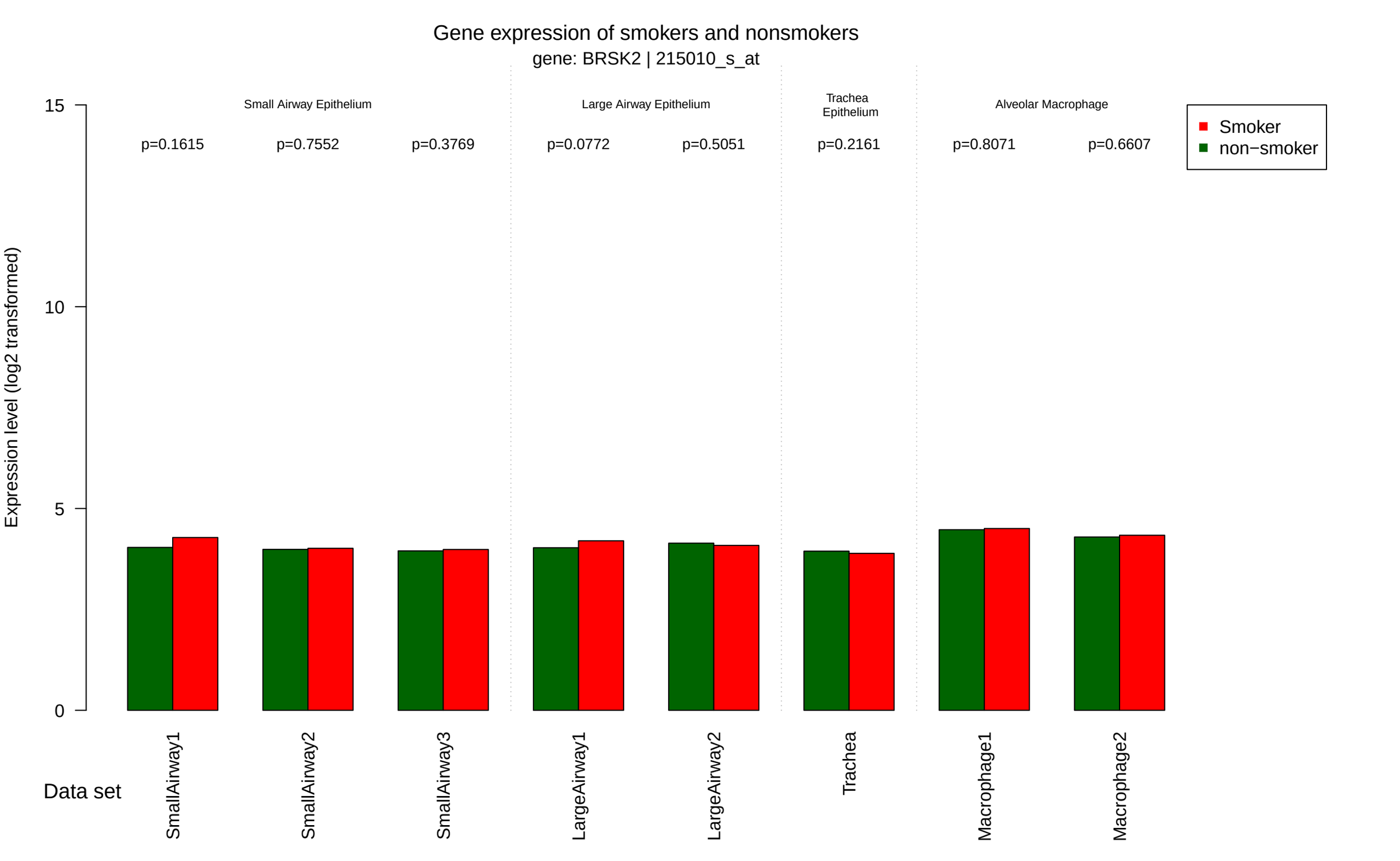
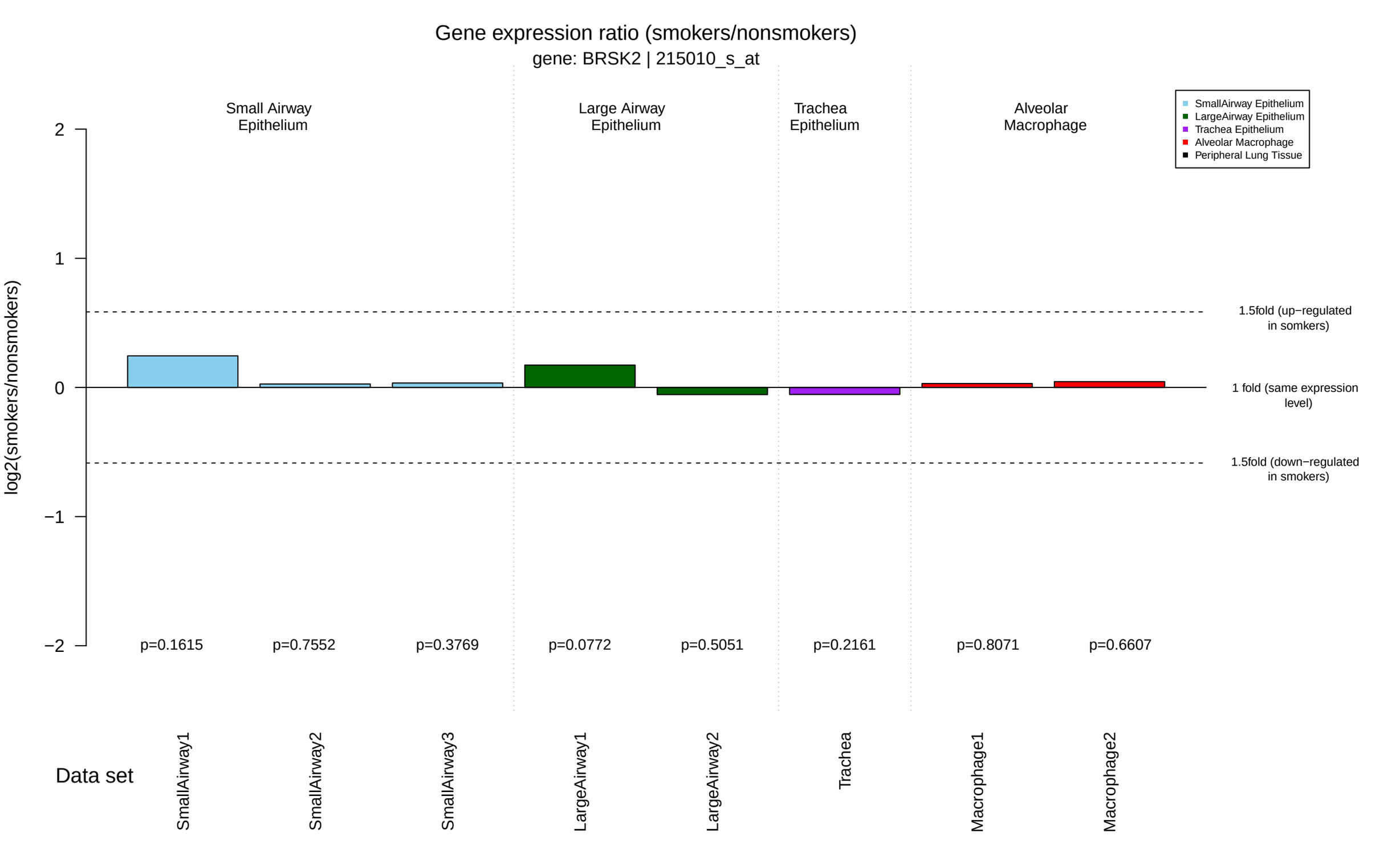
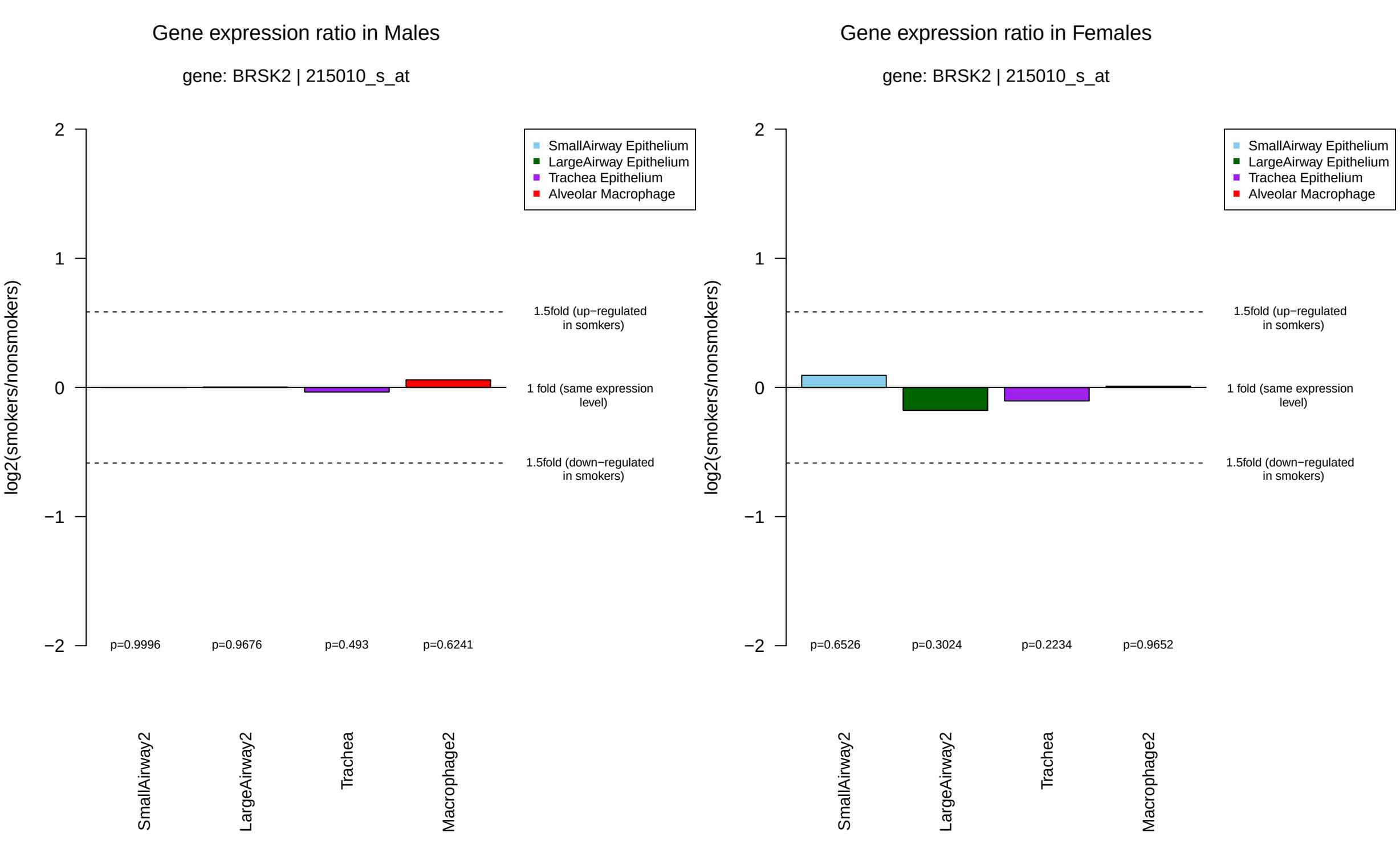
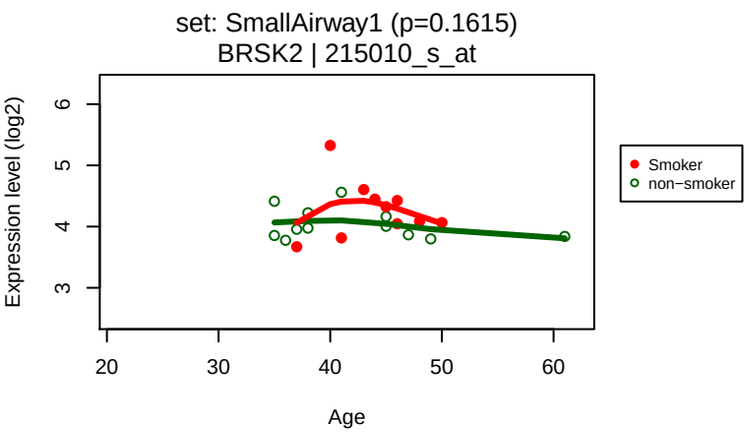
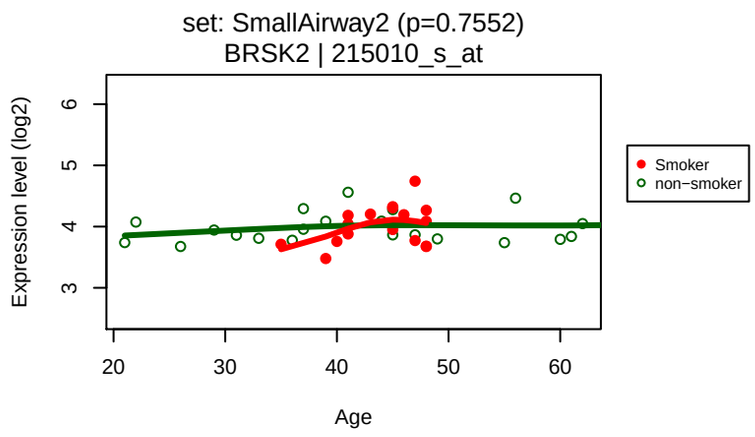
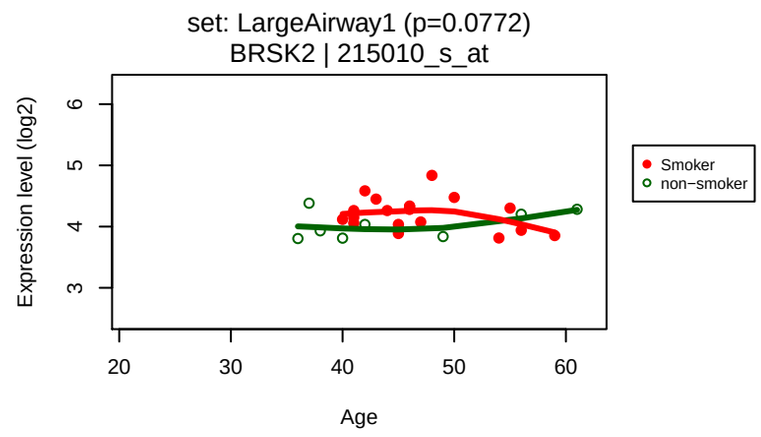

Gene expression of smokers and nonsmokers
gene: BRSK2 | 215010_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1615
p=0.7552
p=0.3769
p=0.0772
p=0.5051
p=0.2161
p=0.8071
p=0.6607
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

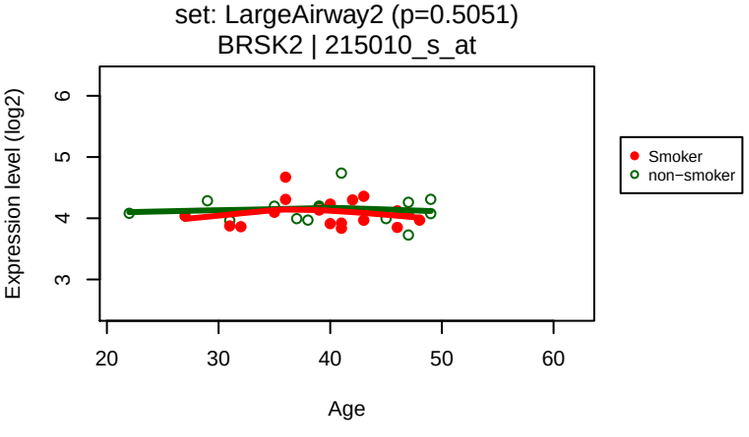

Gene expression ratio (smokers/nonsmokers)
gene: BRSK2 | 215010_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1615
p=0.7552
p=0.3769
p=0.0772
p=0.5051
p=0.2161
p=0.8071
p=0.6607
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

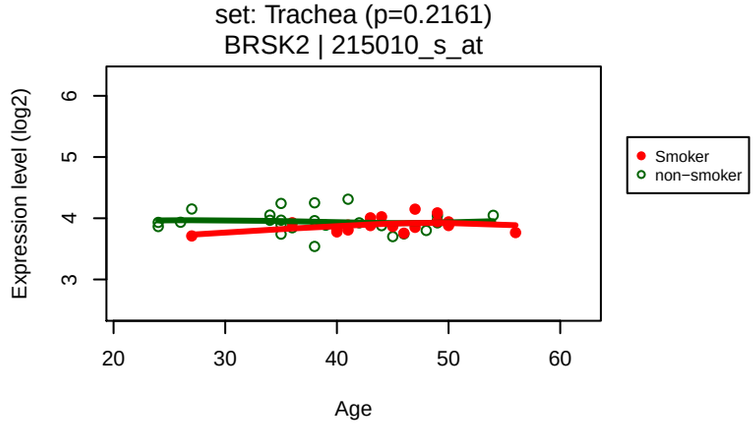

Gene expression ratio in Males
gene: BRSK2 | 215010_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9996
p=0.9676
p=0.493
p=0.6241
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BRSK2 | 215010_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6526
p=0.3024
p=0.2234
p=0.9652
SmallAirway2
LargeAirway2
Trachea
Macrophage2

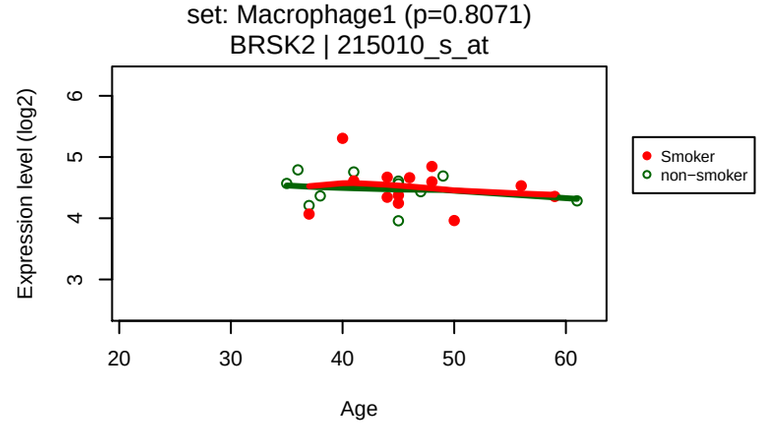

set: SmallAirway1 (p=0.1615)
BRSK2 | 215010_s_at
Expression level (log2)
Age
Smoker
non−smoker

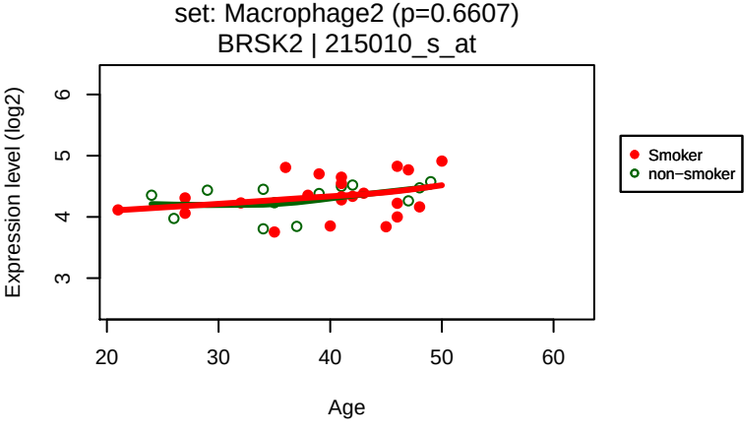

set: SmallAirway2 (p=0.7552)
BRSK2 | 215010_s_at
Expression level (log2)
Age
Smoker
non−smoker

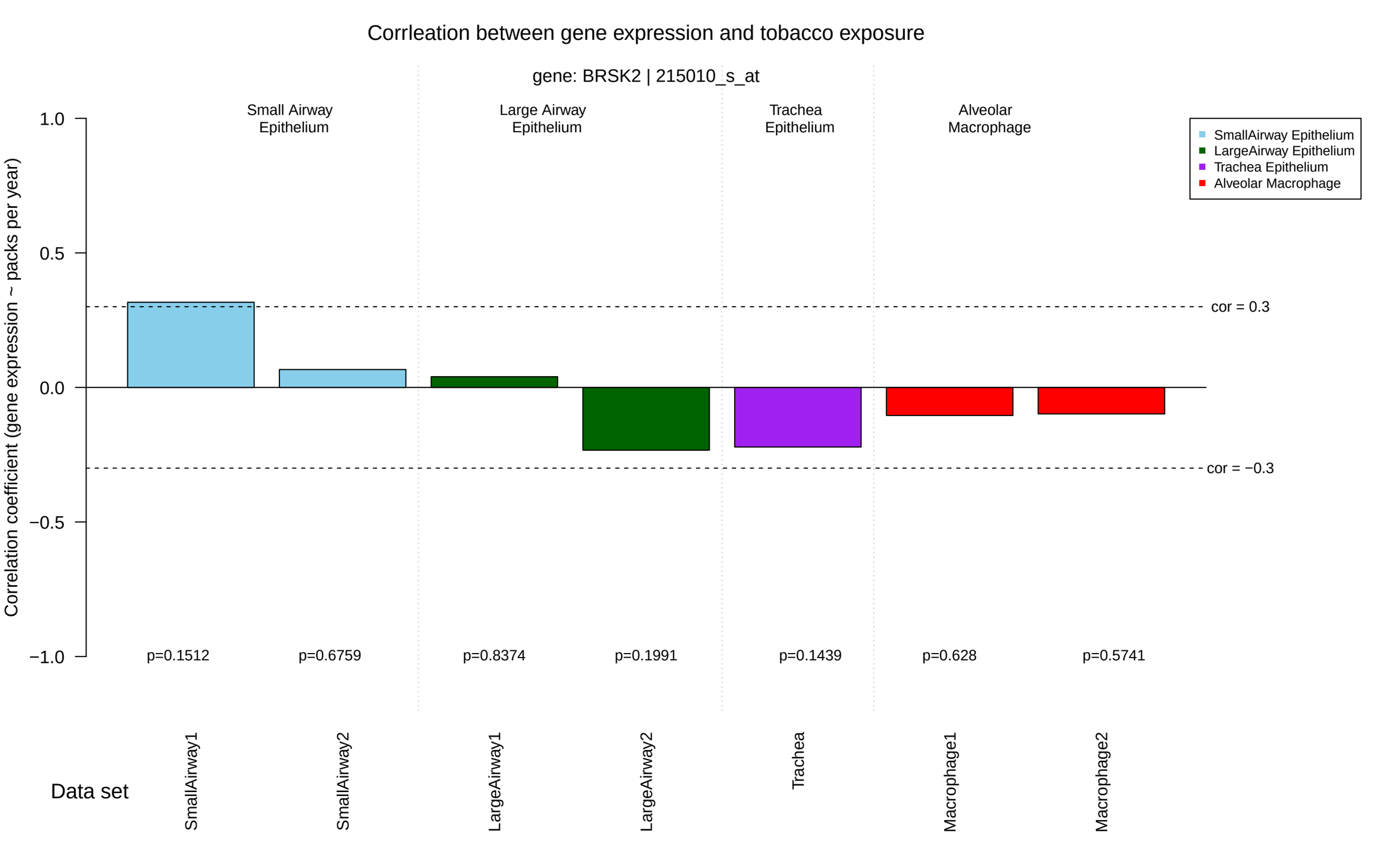

set: LargeAirway1 (p=0.0772)
BRSK2 | 215010_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5051)
BRSK2 | 215010_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2161)
BRSK2 | 215010_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8071)
BRSK2 | 215010_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6607)
BRSK2 | 215010_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BRSK2 | 215010_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1512
p=0.6759
p=0.8374
p=0.1991
p=0.1439
p=0.628
p=0.5741
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2